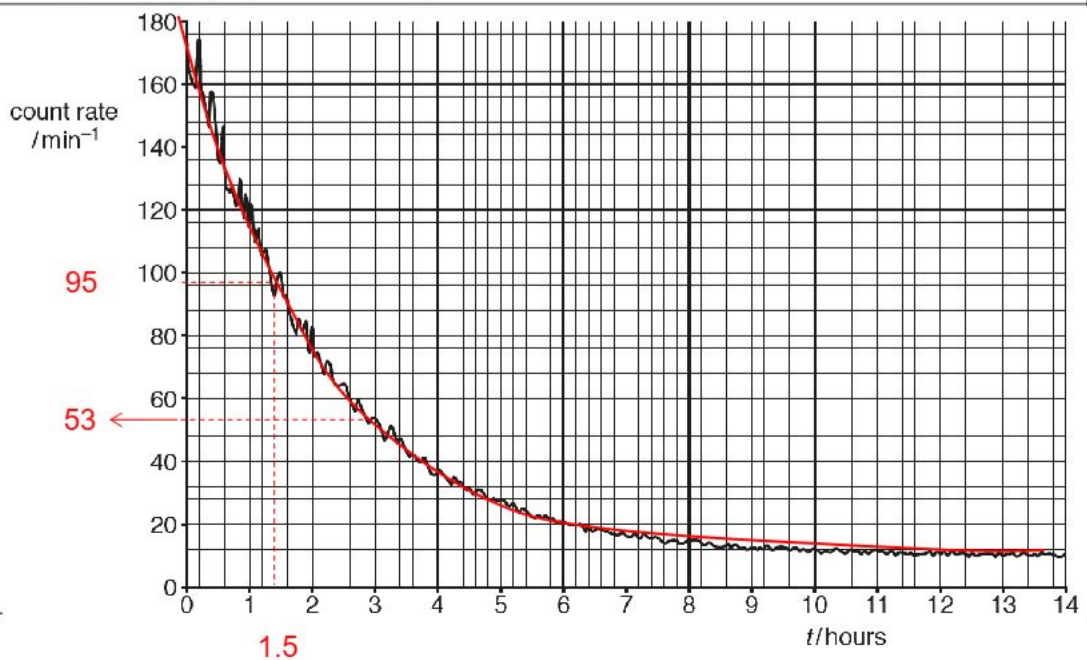
7(a)	Radioactivity (or radioactive decay) is the spontaneous and random disintegration (or decay) of an unstable nucleus into a more stable one (either unstable or become more stable) with the emission of particles (and photons i.e. alpha-particle beta-particle, gamma ray photon). (particles/ radiation – can give marks if mention partial)	**B1** **B1**
7(b)	• emission from radioactive daughter products • self-absorption in source • absorption in air before reaching detector • detector not sensitive to all radiations • window of detector may absorb some radiation • background radiation **(not accepting a repeat mention of part of radiation is detected)** **Any of the above 2 points**	**B1** **B1**
7(c)(i)	Curve is not smooth or curve fluctuates or curve is jagged.	**B1**
7(c)(ii)1.	The curve is almost flat between 10 hours and 14 hours. background radiation = 10 min^{-1} Acceptable range (10 – 12 min^{-1})	**A1**
7(c)(ii)2.		

	half-life determined at least twice Draw a curve of best fit to read off the values from the graph First half-life C_0 = 180 – 10 = 170 min^{-1} (acceptable range: 160 - 170 min^{-1}) $C_1 = \frac{C_0}{2} = \frac{170}{2} = 85\ min^{-1}$ C_1 (measured) = 85 + 10 = 95 min^{-1}, from the graph, $t_{½} = 1.5$ hour (correct method of determining half-life from graph) Second half-life $C_2 = \frac{C_1}{2} = \frac{85}{2} = 43\ min^{-1}$ C_2 (measured) = 43 + 10 = 53 min^{-1}, from the graph, $t_{½} = 3 - 1.5 = 1.5$ hour half-life = 1.5 hours – (mark for background considerations here)	**C1** **C1** **A1**
7(d)	half life : no change because decay is spontaneous / independent of environment	**A1** **B1**